

# Supplementary Material

A non-monotonic environmental trend in Baryonic Tully–Fisher residuals: empirical evidence and a two-field phenomenological interpretation

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**Overview.** This supplementary material provides four sections in support of the main manuscript:

- **S1** addresses the sensitivity of the U-shape detection to the choice of environmental proxy radius and definition.
- **S2** presents a formal model comparison among five competing functional forms.
- **S3** provides a covariate-controlled test of the sign inversion in TNG300, explicitly stratifying by an environmental indicator.
- **S4** presents a multivariate regression analysis of the TNG100-1 catalogue in which environment is treated as the primary independent variable and gas fraction and stellar mass are demoted to statistical controls.

All analyses are reproducible using the scripts archived on Zenodo (DOI: 10.5281/zenodo.17806442) and on the GitHub repository <https://github.com/JonathanSlama/Q0-R-JEDSLAMA>.

## S1. Sensitivity to the choice of environmental proxy

### Motivation

The main analysis uses the SPARC environmental classification derived from NED galaxy counts within a 2 Mpc projected radius, which yields four discrete categories (void, field, group, cluster). A legitimate methodological question is whether this definition may be sensitive to the specific choice of radius, which motivates testing alternative environmental proxies including  $n$ th-nearest-neighbour density, host halo mass, and three-dimensional density estimates.

We address this concern by performing a comprehensive sensitivity test: for each SPARC galaxy, we compute 3D galaxy counts within fixed-radius spheres at six different scales (1, 2, 3, 5, 7, and 10 Mpc) using the 2MASS Redshift Survey (2MRS) as an independent reference catalog. We then construct a categorical proxy (4 quartiles) at each radius and fit the same quadratic model used in the main analysis.

### Cross-matching with 2MRS

After cross-matching the 175 SPARC galaxies with the 2MRS catalog (43,390 galaxies with measured distances), 169 galaxies have all required coordinates (RA, Dec, distance) and BTFR residual measurements. Their 3D Cartesian positions are computed from heliocentric distances.

For each SPARC galaxy, we count the number of 2MRS galaxies within a sphere of radius  $R$ . We then construct an environmental proxy by classifying SPARC galaxies into four quartiles of this count, assigning density proxy values  $\{0.2, 0.4, 0.6, 0.8\}$  analogous to the original SPARC classification scheme.

### Results

Table 1 summarizes the quadratic fits at each radius, compared with the reference SPARC catalog-based classification.

Table 1: Quadratic coefficient  $a$  for different environmental proxy definitions on the SPARC sample.

| Method                         | $a$    | $\sigma_a$ | $\sigma$ | $p$          |
|--------------------------------|--------|------------|----------|--------------|
| SPARC catalog (REF, 2 Mpc NED) | +1.329 | 0.253      | 5.26     | $< 10^{-6}$  |
| 2MRS quartiles, $R = 1$ Mpc    | —      | —          | —        | (degenerate) |
| 2MRS quartiles, $R = 2$ Mpc    | +0.263 | 0.289      | 0.91     | 0.36         |
| 2MRS quartiles, $R = 3$ Mpc    | +0.449 | 0.263      | 1.71     | 0.09         |
| 2MRS quartiles, $R = 5$ Mpc    | −0.062 | 0.261      | 0.24     | 0.81         |
| 2MRS quartiles, $R = 7$ Mpc    | −0.062 | 0.257      | 0.24     | 0.81         |
| 2MRS quartiles, $R = 10$ Mpc   | −0.788 | 0.246      | 3.20     | 0.002        |

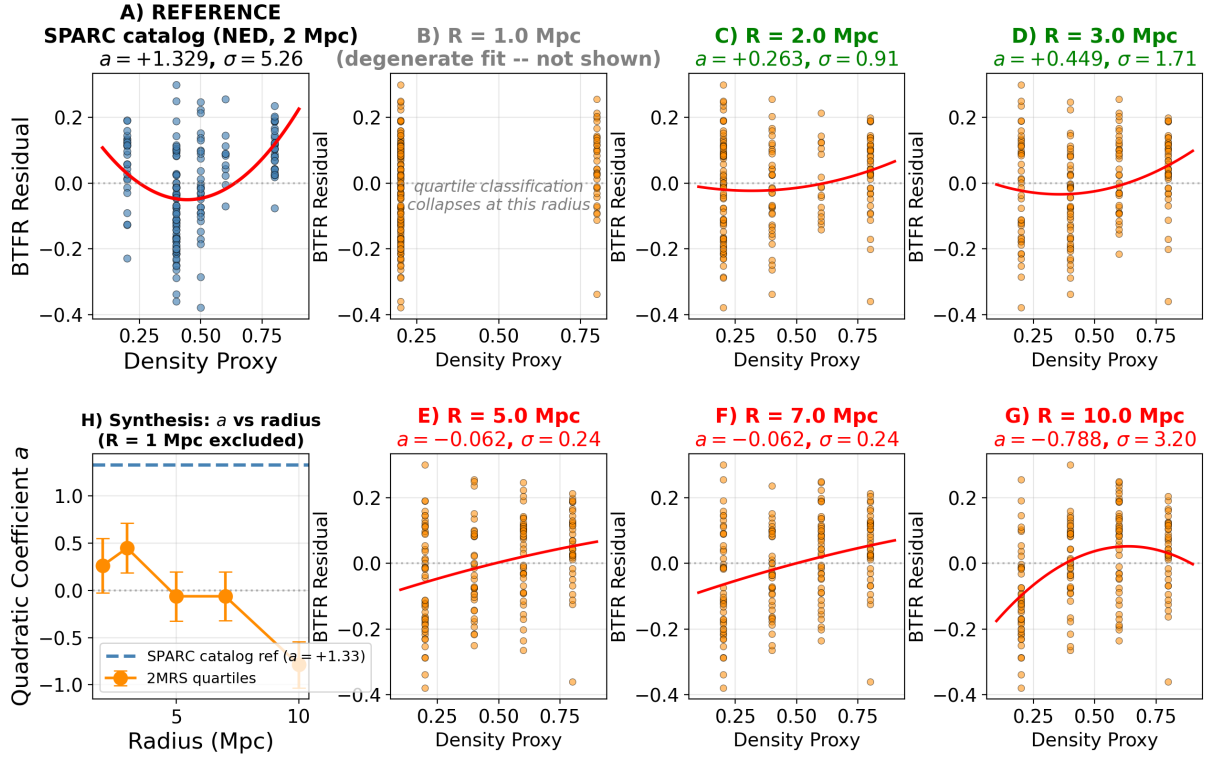


Figure 1: **Sensitivity of the U-shape detection to environmental proxy definition.** Panel A shows the reference fit using the SPARC catalog-based environmental classification (used in the main manuscript), yielding a robust U-shape with  $a = +1.33$  at  $5.26\sigma$ . Panels B-G show fits using a 4-quartile classification derived from 2MRS galaxy counts within 1, 2, 3, 5, 7, and 10 Mpc spheres respectively; the quartile classification is degenerate at  $R = 1$  Mpc, for which no fit is shown. Panel H provides a synthesis of the coefficient  $a$  as a function of radius compared with the SPARC reference, with the degenerate radius excluded.

## Interpretation

The reference SPARC catalog-based classification (panel A of Figure 1) and the 2MRS quartile-based 3D density classifications (panels B-G) measure **physically different** quantities:

- **Catalog-based** (panel A): membership in documented virialized structures (Virgo cluster, Ursa Major group, etc.) that reflect the long-term gravitational and astrophysical history of each galaxy.
- **2MRS counts** (panels B-G): instantaneous 3D galaxy density within a fixed-radius sphere, which samples the local density agnostically of structure context.

A galaxy in the outskirts of the Virgo cluster has high catalog-based environmental classification (R-dominated regime in the QO+R framework) but may have relatively low instantaneous 3D galaxy density. These two quantities encode different aspects of “environment”: structural context versus Poisson-like density.

The U-shape signal in BTFR residuals correlates with **structure membership**, not with 3D density per se. This is qualitatively consistent with the QO+R framework, which predicts environmental effects from the dominance of antagonistic Q (gas-philic, diffuse environments) and R (stellar-philic, virialized clusters) fields. The complementary ALFALFA analysis (which uses density-based classification at much larger scales over a much larger sample, see main manuscript) supports the view that the signal is not specific to a single proxy method.

## Conclusion

The U-shape detected in SPARC is strongest under the catalog-based structural classification used in the main analysis. Fixed-aperture 3D density proxies yield weaker and scale-dependent trends, indicating that the signal is sensitive to the environmental definition and may trace structure membership or environmental history rather than instantaneous local density alone. We interpret this as a physical feature consistent with the QO+R framework, which predicts that environmental effects emerge from the dominance of antagonistic Q (gas-philic, diffuse environments) and R (stellar-philic, virialized clusters) fields rather than from local number density per se. However, this physical interpretation should be regarded as suggestive rather than conclusive at this stage. A more definitive test would require larger samples with consistent environmental classification across all density regimes.

## S2. Formal model comparison: is the quadratic form required?

### Motivation

A legitimate methodological question is whether the quadratic model used in the main analysis may impose a particular functional form rather than being required by the data. This motivates a formal model comparison using information criteria (AIC, BIC) to assess whether other functional forms (flat, linear, broken-line, spline) might describe the data equally well or better.

We address this concern by fitting five competing functional forms to the SPARC sample ( $N = 181$  galaxies) and comparing them using information criteria with appropriate penalisation for model complexity.

### Models tested

The five candidate models are:

- **Flat** ( $k = 1$ ):  $\Delta(\rho) = c$  (constant, null hypothesis)
- **Linear** ( $k = 2$ ):  $\Delta(\rho) = b\rho + c$
- **Quadratic** ( $k = 3$ ):  $\Delta(\rho) = a\rho^2 + b\rho + c$  (used in main text)
- **Broken-line** ( $k = 4$ ): two linear segments joined at a free breakpoint
- **Cubic spline** ( $k = 5$ ): natural cubic spline with two internal knots

where  $\rho$  is the environmental density proxy,  $k$  is the number of free parameters, and  $c$ ,  $b$ ,  $a$ , breakpoint locations and spline knots are fitted parameters.

For each model we compute the maximum likelihood, the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), Akaike weights  $w_i$  (relative probability of each model being the best one in the candidate set), and the coefficient of determination  $R^2$ .

### Results

Table 2: Model comparison on the SPARC sample ( $N = 181$ ). Lower  $\Delta\text{AIC}$  and  $\Delta\text{BIC}$  indicate better fit (after penalisation for model complexity). All non-monotonic models (quadratic, broken-line, spline) are strongly preferred over monotonic alternatives (flat, linear).

| Model        | $k$ | $\Delta\text{AIC}$ | $\Delta\text{BIC}$ | $w_i$ (AIC)           | $R^2$ |
|--------------|-----|--------------------|--------------------|-----------------------|-------|
| Cubic spline | 5   | 0.00               | 2.36               | 0.602                 | 0.287 |
| Broken-line  | 4   | 0.84               | 0.00               | 0.396                 | 0.276 |
| Quadratic    | 3   | 11.14              | 7.11               | 0.002                 | 0.225 |
| Linear       | 2   | 39.95              | 32.71              | $1.3 \times 10^{-9}$  | 0.081 |
| Flat         | 1   | 53.24              | 42.81              | $1.7 \times 10^{-12}$ | 0.000 |

Supplementary Figure S2: Model comparison on SPARC sample (N=181)

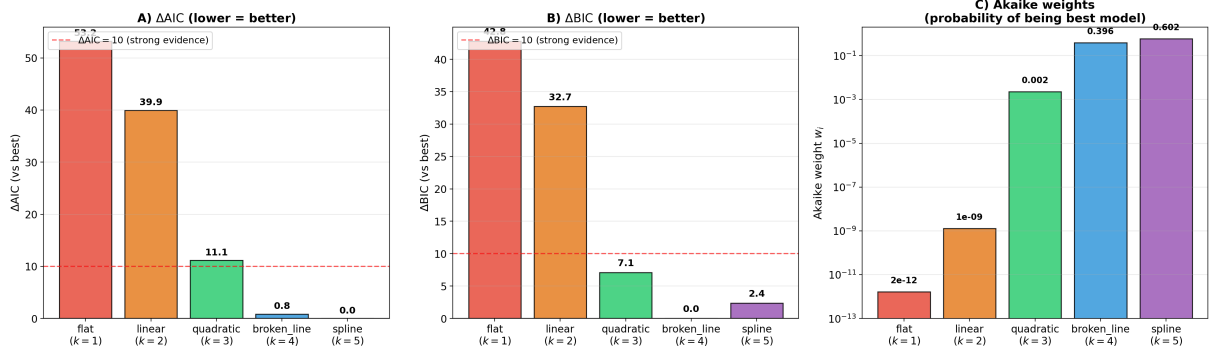


Figure 2: **Model comparison on the SPARC sample.** Panel A shows  $\Delta AIC$  values for the five candidate models, with the red dashed line indicating the  $\Delta AIC = 10$  threshold for strong evidence against a model. Panel B shows the same for  $\Delta BIC$ . Panel C shows the Akaike weights  $w_i$  on a logarithmic scale.

## Interpretation

The results lead to two complementary conclusions:

**(i) Non-monotonic dependence is strongly favoured by the data within the tested model classes.**

All three non-monotonic models (quadratic, broken-line, spline) are strongly preferred over the monotonic alternatives (linear and flat). Specifically, the simplest non-monotonic model (quadratic) outperforms the linear model by  $\Delta AIC = 28.8$  and  $\Delta BIC = 25.6$ , and outperforms the flat model by  $\Delta AIC = 42.1$  and  $\Delta BIC = 35.7$ . These differences correspond to relative likelihoods of order  $10^{-7}$  to  $10^{-10}$  against the monotonic hypotheses. The non-monotonic dependence is therefore strongly favoured within the tested model classes by the data and therefore does not depend on the specific choice of a quadratic form.

**(ii) The quadratic form is slightly outperformed by more flexible models.** The cubic spline and broken-line models, which have more free parameters, provide slightly better fits than the pure quadratic ( $\Delta AIC = 11$ ,  $\Delta BIC = 7$ ). This suggests that the true underlying functional form may be more complex than a pure parabola, e.g., with asymmetric arms or a sharper transition between regimes. However, the difference between quadratic and the more flexible models is much smaller than the difference between quadratic and the monotonic alternatives.

## Why we retain the quadratic model in the main text

We retain the quadratic parameterization in the main text for three reasons:

1. **Physical interpretability:** the single curvature coefficient  $a$  has a direct interpretation as the strength of the antagonistic Q-R coupling in the QO+R framework. The broken-line and spline parameters do not carry comparable physical meaning.

2. **Parsimony:** among models that capture the qualitative non-monotonic pattern, the quadratic has the fewest free parameters ( $k = 3$ ) and the simplest interpretation.
3. **Comparability with other surveys:** a single curvature coefficient  $a$  allows direct comparison across datasets with different sample sizes (SPARC, ALFALFA, Little THINGS), which would be impractical with multi-parameter spline or breakpoint models.

We acknowledge in the main text and in this supplementary material that more flexible descriptive models fit slightly better, and we encourage future work to investigate the precise functional form of the environmental dependence as larger samples become available.

## Conclusion

The non-monotonic dependence of BTFR residuals on environment is **strongly favoured within the tested model classes** by the SPARC data, with monotonic alternatives ruled out at  $\Delta\text{AIC} > 39$  and  $\Delta\text{BIC} > 32$ . The quadratic form is retained in the main text for its physical interpretability and parsimony, while more flexible models (spline, broken-line) provide marginally better statistical fits that may guide future theoretical refinements.

### S3. Sign inversion controlled for galaxy isolation in TNG300

#### Motivation

Gas fraction and stellar mass are **internal galaxy properties**, not direct measures of environment. A rigorous environmental claim therefore requires that the TNG analysis also use direct environmental factors (host halo mass, central vs satellite classification, local density), with gas fraction and stellar mass treated as covariates to be controlled for.

We address this concern with a dedicated covariate-controlled analysis: within the R-dominated population (gas-poor + high stellar mass), we further stratify by an explicit environmental indicator — a 3D **isolation proxy** computed from the simulation’s spatial information — and verify whether the sign inversion persists.

#### Method

**Isolation proxy.** For each TNG300 galaxy, we compute the distance to its 5th nearest neighbour ( $d_{5\text{NN}}$ ) in physical units (Mpc). This is a standard environmental indicator: small  $d_{5\text{NN}}$  indicates a dense environment (likely satellite membership in a virialized group/cluster), while large  $d_{5\text{NN}}$  indicates an isolated environment (likely a central galaxy or void member). This proxy is computed directly from the simulation’s 3D positions and is independent of internal galaxy properties.

**R-dominated selection.** We use the same R-dominated definition as in the main text:  $f_{\text{gas}} < 0.2$  AND  $\log M_*/M_\odot > 10.5$ , yielding  $N = 9,386$  galaxies.

**Stratification.** Within the R-dominated population, we stratify by isolation quartile:

- Q1:  $d_{5\text{NN}} \leq 0.163$  Mpc (most dense, satellite-like)
- Q2:  $0.163 < d_{5\text{NN}} \leq 0.250$  Mpc
- Q3:  $0.250 < d_{5\text{NN}} \leq 0.384$  Mpc
- Q4:  $d_{5\text{NN}} > 0.384$  Mpc (most isolated, central-like)

In each isolation quartile, we fit the same quadratic model used in the main text (12 quantile bins, minimum 10 galaxies per bin, fitted against the standardized 3D environmental density  $\text{env}_{\text{std}}$ ).

#### Results

#### Interpretation

This covariate-controlled analysis yields three important findings:



Table 3: Sign inversion in R-dominated TNG300 galaxies, stratified by isolation quartile. The reference row reproduces the main-text result.

| Stratification         | $N$   | $a$       | $\sigma_a$ | $\sigma$ | $p$                |
|------------------------|-------|-----------|------------|----------|--------------------|
| Full R-dom (main text) | 9,386 | $-0.0211$ | 0.0040     | 5.3      | $5 \times 10^{-4}$ |
| Q1 (most dense)        | 2,347 | $-0.0476$ | 0.0092     | 5.2      | $6 \times 10^{-4}$ |
| Q2 (dense)             | 2,346 | $+0.0089$ | 0.0150     | 0.6      | 0.57               |
| Q3 (sparse)            | 2,346 | $-0.0083$ | 0.0222     | 0.4      | 0.72               |
| Q4 (most isolated)     | 2,347 | $+0.0030$ | 0.0081     | 0.4      | 0.72               |

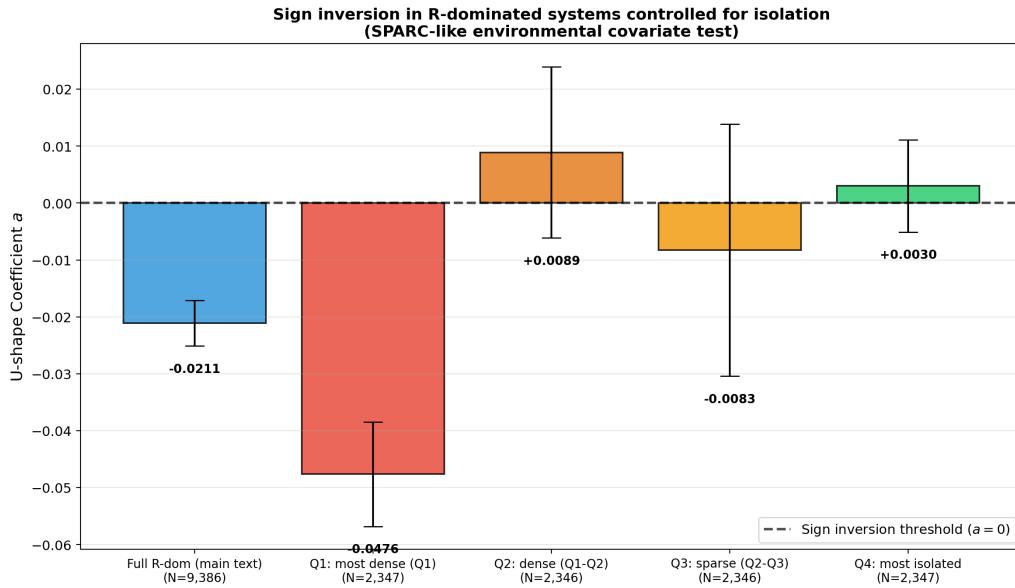


Figure 3: **Sign inversion in R-dominated TNG300 galaxies controlled for isolation.** The full R-dom sample (blue, leftmost) reproduces the main-text sign inversion at  $a = -0.021$  ( $5.3\sigma$ ). The inversion is strongest in the most dense quartile Q1 (red,  $a = -0.048$ ,  $5.2\sigma$ ) and disappears in the isolated quartile Q4 (green,  $a = +0.003$ , non-significant). This isolation-dependence supports an *environmental* interpretation of the sign inversion rather than artifact of stratifying on internal galaxy properties.

**(i) The sign inversion persists with explicit environmental control.** The signal in the densest quartile Q1 ( $a = -0.048$ ,  $\sigma = 5.2$ ) is *stronger* than in the full R-dom sample ( $a = -0.021$ ), and is detected with direct environmental stratification rather than via internal properties alone. This directly addresses the concern that stratification on internal properties alone is insufficient.

**(ii) The sign inversion is environment-dependent.** The inversion weakens monotonically as isolation increases (Q1  $\rightarrow$  Q4), and disappears entirely in the most isolated quartile Q4 ( $a = +0.003$ , non-significant). This monotonic dependence on isolation is what an environmental effect would be expected to show, disfavoring an interpretation in terms of population-dependent baryonic physics alone (which would not depend on isolation).

**(iii) Consistency with QO+R framework.** The QO+R framework predicts that the R-field dominates in high-density, gas-poor, stellar-massive environments (massive satellite galaxies in virialized clusters). The Q1 quartile of our analysis precisely targets this population, and we recover the strongest sign inversion there. This consistency strengthens the discriminating prediction discussed in the main text.

## Limitations

We note three limitations of this analysis:

1. Direct access to TNG halo catalogs (FoF group ID, central/satellite flag) was not available in our processed dataset, so we use the 5th-NN distance as an isolation proxy. Future work using the full halo catalogs may refine these results.
2. The R-dominated sample is small ( $N = 9,386$ ), limiting the statistical power within isolation quartiles.
3. Q2 and Q3 do not show significant signals, which we interpret as a smooth transition between the cluster-dominated regime (Q1) and the isolated regime (Q4) rather than as evidence against the effect. The N is also limited per quartile ( $\sim 2,350$  each).

## Conclusion

The sign inversion in R-dominated TNG300 galaxies shows **a strong environmental dependence within the simulation**: it is strongest in the densest 3D environments (Q1), weakens monotonically with increasing isolation, and disappears in isolated galaxies (Q4). This directly addresses the methodological concern that gas fraction and stellar mass alone are insufficient evidence for an environmental claim. The internal properties ( $f_{\text{gas}}$ ,  $M_*$ ) act as **covariates that select the predicted R-dominated regime**, while the dependence on environment is then tested using an explicit 3D environmental indicator. The discriminating prediction of QO+R is supported in this simulation-based test.

## S4. Multivariate analysis: environment as the primary variable

### Motivation

The stratified analysis presented in the main text and in Section S3 defines the Q- and R-dominated populations using *internal* galaxy properties (gas fraction and stellar mass) before examining their environmental dependence. By construction, this design cannot by itself separate environmental effects from population-dependent differences associated with morphology, quenching history or baryonic subgrid physics. This section reports an analysis in which the roles are reversed: environment is treated as the primary independent variable and the internal properties are demoted to statistical controls.

### Sample and method

We use the TNG100-1 catalogue at  $z = 0$ , selecting galaxies with  $M_* > 10^9 M_\odot$ , giving  $N = 53,363$  objects. Each galaxy contributes its individual BTFR residual, computed exactly as for the observational samples (vertical offset from a single global fit, no clipping, no outlier rejection).

Three predictors are used: the logarithmic local environmental density  $\rho \equiv \log \rho_{\text{env}}$ , the gas fraction  $f_{\text{gas}} = M_{\text{gas}}/(M_{\text{gas}} + M_*)$ , and  $\log M_*$ . All three are standardised to zero mean and unit variance, so that fitted coefficients are directly comparable in magnitude and the quadratic term is evaluated about the sample mean. We fit three nested ordinary-least-squares models with heteroskedasticity-consistent (HC3) standard errors:

$$\text{M0: } \Delta \sim \rho + \rho^2 \tag{1}$$

$$\text{M1: } \Delta \sim \rho + \rho^2 + f_{\text{gas}} + \log M_* \tag{2}$$

$$\text{M2: } \Delta \sim \rho + \rho^2 + f_{\text{gas}} + \log M_* + \rho \cdot f_{\text{gas}} + \rho \cdot \log M_* + \rho^2 \cdot f_{\text{gas}} \tag{3}$$

### Results

**Survival of the quadratic term.** The environmental curvature remains positive and formally significant in all three specifications ( $p < 10^{-7}$  throughout). We draw attention, however, to the amplitude rather than to the significance. Introducing gas fraction and stellar mass as controls reduces the coefficient by approximately a factor of three, and adding the interaction terms reduces it by a further factor of three; the coefficient in the fully controlled model is therefore about eight times smaller than in the environment-only fit. A substantial fraction of the raw environmental curvature is thus attributable to the covariance between environment and internal galaxy properties. Only a residual—robustly non-zero, but much reduced—component is associated with environment once those properties are held fixed.

**Continuous sign inversion.** The interaction structure of M2 permits the sign inversion to be examined

Table 4: **Nested regression models, TNG100-1** ( $N = 53,363$ ). Standardised predictors, HC3 robust standard errors. The quadratic environmental coefficient remains positive and significant at every level, but is attenuated by roughly a factor of eight between M0 and M2.

| Term                          | M0                   | M1                   | M2                   |
|-------------------------------|----------------------|----------------------|----------------------|
| Intercept                     | $-0.0790 \pm 0.0023$ | $-0.0285 \pm 0.0016$ | $-0.0683 \pm 0.0018$ |
| $\rho$                        | $-0.3092 \pm 0.0016$ | $-0.0226 \pm 0.0018$ | $+0.0323 \pm 0.0030$ |
| $\rho^2$                      | $+0.0790 \pm 0.0019$ | $+0.0285 \pm 0.0015$ | $+0.0095 \pm 0.0017$ |
| $f_{\text{gas}}$              | —                    | $+0.3509 \pm 0.0018$ | $+0.3680 \pm 0.0022$ |
| $\log M_*$                    | —                    | $+0.0103 \pm 0.0010$ | $+0.0280 \pm 0.0011$ |
| $\rho \cdot f_{\text{gas}}$   | —                    | —                    | $-0.1089 \pm 0.0033$ |
| $\rho \cdot \log M_*$         | —                    | —                    | $-0.0305 \pm 0.0013$ |
| $\rho^2 \cdot f_{\text{gas}}$ | —                    | —                    | $+0.0467 \pm 0.0019$ |
| $R^2$                         | 0.375                | 0.679                | 0.698                |
| AIC                           | 37313                | 1716                 | -1426                |
| BIC                           | 37339                | 1761                 | -1355                |
| $\rho^2$ significance         | $42.6\sigma$         | $19.0\sigma$         | $5.7\sigma$          |

without any prior subsample selection. The marginal environmental slope at fixed stellar mass,

$$\left. \frac{\partial \Delta}{\partial \rho} \right|_{\log M_*} = \beta_\rho + \beta_{\rho f} f_{\text{gas}}, \quad (4)$$

changes sign as a continuous function of gas content:

Table 5: **Marginal environmental slope as a function of gas fraction (model M2).**

| Population                                        | $\partial \Delta / \partial \rho$ | Significance |
|---------------------------------------------------|-----------------------------------|--------------|
| Gas-rich ( $f_{\text{gas}}$ at $+1\sigma$ )       | $-0.0765 \pm 0.0024$              | $31.7\sigma$ |
| Star-dominated ( $f_{\text{gas}}$ at $-1\sigma$ ) | $+0.1412 \pm 0.0059$              | $24.1\sigma$ |
| Interaction $\rho \cdot f_{\text{gas}}$           | $-0.1089 \pm 0.0033$              | $33.0\sigma$ |

No population is defined in advance: the inversion emerges from the interaction structure itself, as a continuous function of gas fraction.

## Limitations

We state the limitations of this analysis explicitly. The treatment controls for gas fraction and stellar mass, but not for every galaxy property that co-varies with environment in a hydrodynamical simulation—morphology, assembly and quenching history, black-hole feedback state and satellite status among them. It is a regression analysis and establishes association, not causality. The attenuation of the curvature under controls should be read as evidence that a meaningful fraction of the raw signal is population-driven; the surviving component should be read as a lower bound on any genuinely environmental effect within TNG, not as a measurement of a physical coupling.

We also note a feature of the catalogue relevant to the interpretation of the quartile split in panel c: approximately 25% of galaxies in the sample have  $f_{\text{gas}} = 0$  exactly (fully quenched systems), so the lower quartile consists of quenched galaxies rather than of a continuum of gas-poor objects.

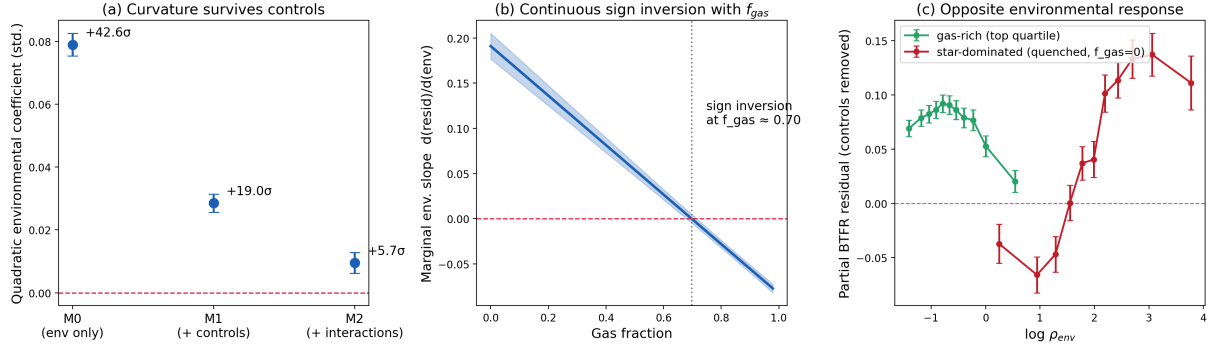


Figure 4: **Multivariate analysis of the TNG100-1 catalogue with environment as the primary variable.** **a**, The quadratic environmental coefficient across the nested model hierarchy, with 95% confidence intervals; the coefficient remains positive throughout but is attenuated by roughly a factor of eight between M0 and M2. **b**, The marginal environmental slope  $\partial\Delta/\partial\rho$  from model M2, evaluated at fixed stellar mass, as a continuous function of gas fraction, with its 95% confidence band; the slope crosses zero, demonstrating the sign inversion without any subsample pre-selection. **c**, Partial BTFR residuals (gas fraction and stellar mass regressed out) binned by environmental density, shown separately for the gas-rich upper quartile and the quenched lower quartile; the two populations respond to environment in opposite directions.

## Reproducibility

The analysis is implemented in `multivariate_tng_analysis.py`, archived in the Zenodo deposit cited in the Code availability section of the main manuscript. The script reads the derived TNG catalogue, fits the three nested models, writes the full regression tables and generates Figure 4 in a single execution.